

Statistics 6910, Section 003, Fall 2013, Quiz 2 (25 Points)

Wednesday, September 18, 2013

Your Name: _____

Question 1: Vectors (15 Points)

The vector `examScores` is created as follows in R:

```
> examScores = c(Adam = 10, Cathy = 18, Emma = 15, Paul = 18, Steve = 12)
```

```
> examScores
```

Adam	Cathy	Emma	Paul	Steve
10	18	15	18	12

For each of the following 10 blocks of R expressions, indicate the resulting output. Note that some of the expressions change the content of `examScores` while others leave it unchanged. You should assume that these expressions are **evaluated sequentially**. So, if `examScores` was modified via a previous expression, you need to proceed with the modified content of `examScores`, and not with the original content. If something results in an error message, state **Error**. Read carefully! Clearly indicate whether the names appear in the output or not! (1.5 points each)

#1

```
examScores[examScores == max(examScores)]
```

#2 (score adjusted!)

```
examScores[-(1:4)] = 13
```

```
examScores
```

#3

```
examScores[c("Adam", "Paul")]
```

#4

```
examScores[Emma]
```

#5

```
examScores > 16
```

#6

```
examScores[examScores > 16]
```

#7 (someone cheated - score is invalid)

```
examScores[2] = NA
```

```
examScores
```

#8 (include the maximum in this vector)

```
examScores[7] = 20
```

```
examScores
```

#9 (be careful !)

```
mean(examScores)
```

#10 (next exam)

```
examScores = 0
```

```
examScores
```

Question 2: Sequences (10 Points)

Use `rep()`, `seq()`, `c()`, or `:` for the following tasks. As you have no access to help files, I would suggest that you specify arguments by name — unless you are absolutely certain which argument appears in which position. **Be as efficient as possible.** (2 points each)

- Create a vector with values:

```
[1] 10  9  8  7  6  5  4  3  2  1  1  2  3  4  5  6  7  8  9 10
```

- Create a vector with values:

```
[1]  0  0  0  0  0  0 25 25 25 25 50 50 50 75 75 100
```

- Create a vector with values (count carefully!):

```
[1] 1.000000 2.166667 3.333333 4.500000 5.666667 6.833333 8.000000
```

- Create a vector with values:

```
[1]  0  5 10  0  5 10  0  5 10  0  5 10  0  5 10
```

- The following expressions are executed in R. What will be the most likely content of the vector `pp`?

```
library(gmp) # a package for Multiple Precision Arithmetic

xx = isprime(1:10)
pp = (1:10)[xx > 0]
pp
```

The help page for `isprime` indicates:

Determine if number is (very probably) prime

Description

Determine whether the number `n` is prime or not, with three possible answers:

2:
`n`
 is prime,

1:
`n`
 is probably prime (without being certain),

0:
`n`
 is composite.